

2023 AMC 10A Problem 5 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 5. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 5

Problem:

How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?

Answer Choices:

- A. 14
- B. 15
- C. 16
- D. 17
- E. 18

Solution:

Because $8 = 2^3$ and $15 = 3 \cdot 5$, the prime factorization of the given number is $2^{15} \cdot 3^5 \cdot 5^{15}$, which is $3^5 \cdot 10^{15} = 243 \cdot 10^{15}$. The base-ten representation is 243 followed by 15 zeros, so it has (E) 18 digits in all.

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